WTW 158 - MEMO - **WORKSHEET ON DIFFERENTIATION -2016**

Find the derivatives of the following functions.

1. $\frac{d}{dx}x^2e^{4x} = 2x \times e^{4x} + x^2 \times 4e^{4x}$

2. $\frac{d}{d\theta}\frac{\tan 2\theta}{1+\sec(\theta^2)} = \frac{2\sec^2 2\theta[1+\sec(\theta^2)] - \tan 2\theta \sec(\theta^2)\tan(\theta^2)2\theta}{[1+\sec(\theta^2)]^2}$

3. $\frac{d}{dt}(t^3 + e^t + 3^t) = 3t^2 + e^t + 3^t \ln 3$

4. $\frac{d}{dt}[(1-4t)^3 + e^{1-5t} + 3^{1-6t}] = 3(1-4t)^2(-4) + e^{1-5t}(-5) + (\ln 3)3^{1-6t}(-6)$

5. $\frac{d}{dx}5\sin(x^4) = 5 \times \cos(x^4) \times 4x^3$

6. $\frac{d}{dx}\sin^4 5x = 4\sin^3(5x) \times \cos(5x) \times 5$

7. $\frac{d}{dt}e^{\cos 5t} = e^{\cos 5t} \times -\sin 5t \times 5$

8. $\frac{d}{dt}\cos(e^{5t}) = -\sin(e^{5t}) \times e^{5t} \times 5$

9. $\frac{d}{du}\ln(1-\sqrt[3]{u}) = \frac{1}{1-\sqrt[3]{u}} \times -\tfrac{1}{3}u^{-2/3}$

10. $\frac{d}{du}\sqrt[3]{1-\ln 2u} = \tfrac{1}{3}(1-\ln 2u)^{-2/3} \times \frac{2}{2u}$

11. $\frac{d}{dx}\sec(\ln(2x+6)) = \sec(\ln(2x+6))\tan(\ln(2x+6)) \times \frac{2}{2x+6}$

12. $\frac{d}{dt}\ln(\sec(3x+7)) = \frac{1}{\sec(3x+7)} \times \sec(3x+7)\tan(3x+7) \times 3$

13. $\frac{d}{dx}\operatorname{cosec}(3x^2) = -\operatorname{cosec}(3x^2)\cot(3x^2) \times 6x$

14. $\frac{d}{dx}3^{\operatorname{cosec}x^2} = (\ln 3)3^{\operatorname{cosec}x^2} \times -\operatorname{cosec}(x^2)\cot(x^2) \times 2x$

15. $\frac{d}{dx}\tan^{-1}(e^{8x}) = \frac{8e^{8x}}{1+(e^{8x})^2}$

16. $\frac{d}{dt}\sin^{-1}(1-9t^4) = \frac{1}{\sqrt{1-(1-9t^4)^2}} \times -36t^3$

17. $x^3\sin y + xy^4 = y + e \Rightarrow$:

$3x^2\sin(y) + x^3\cos(y)y' + y^4 + 4xy^3y' = y' \Rightarrow y' = -\frac{3x^2\sin y + y^4}{x^3\cos y + 4xy^3 - 1}$

18. $f(t) = t^{\cos 2t} \Rightarrow \ln f(t) = \ln t^{\cos 2t} = \cos 2t \ln t$

$\Rightarrow \frac{d}{dt}\ln f(t) = \frac{d}{dt}[\cos 2t \ln t]$

$\Rightarrow \frac{f'(x)}{f(x)} = -2\sin 2t \times \ln t + \cos 2t \times \frac{1}{t}$

$\Rightarrow f'(x) = f(x)\left[-2\sin 2t \times \ln t + \cos 2t \times \frac{1}{t}\right]$

19. $f(x) = 2e^{1-x} \Rightarrow f'(x) = -2e^{1-x}$

$\Rightarrow f(1) = 2$ and $f'(1) = -2$

The equation is $y - f(1) = f'(1)(x-2)$

$\Rightarrow y - 1 = -2(x-2)$

$\Rightarrow y = -2x + 5$

20. $f'(x) = \frac{d}{dx}\frac{\ln 2x}{x^3} = \frac{\frac{2}{2x} \times x^3 - \ln 2x \times 3x^2}{(x^3)^2} = \frac{x^2 - \ln 2x \times 3x^2}{x^6} = \frac{1 - 3\ln 2x}{x^4}$

$f'(x) = 0 \Rightarrow 1 - 3\ln(2x) = 0 \Rightarrow \ln(2x) = \frac{1}{3} \Rightarrow 2x = e^{1/3} \Rightarrow x = 0.5e^{1/3}$